



Intercostal nerve cryoablation with nerve blocks reduces postoperative medication use after minimally invasive pediatric pectus excavatum repair

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Received: 22 May 2025 / Accepted: 12 January 2026

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Abstract

Purpose Minimally invasive pectus excavatum repair (MIRPE) using the Nuss procedure is associated with significant postoperative pain. Intercostal nerve cryoablation (INC), used alone or with adjuncts like intercostal nerve blocks (INB) have improved pain control. This retrospective study aimed to compare the impact of INC with INB on overall postoperative medication exposure.

Methods Patients aged ≤20 years who underwent MIRPE with bilateral INC and INB (INC-INB) from 2021 - March 2023 were compared to historical controls with bilateral paravertebral blocks (PVB) from 2018 to 2020. Demographics, operative details, and outcomes were compared using Chi-square and Kruskal-Wallis tests.

Results Of 164 patients, 90 (55%) underwent repair with INC-INB, and 74 (45%) with PVB. Age, Haller index, and second bar usage were comparable between groups. While median (IQR) operative time was longer for patients with INC-INB [110 (102, 122) vs. 69 (60, 82) minutes, $p < .001$], total operating room time was equivalent [185 (169, 199) vs. 194 (174, 209) minutes, $p = .09$]. Compared to PVB, INC-INB was associated with lower opioid consumption [0 (0, 0.2) vs. 1.6 (0.7, 3.0) OME/kg, $p < .001$], fewer total postoperative medications [6 (4, 7) vs. 12 (11, 13), $p < .001$], and shorter length of stay [30 (28, 34) vs. 59 (55, 77) hours, $p < .001$].

Conclusion INC-INB reduces overall postoperative medication exposure and length of stay compared to percutaneous PVB for pain control in children undergoing MIRPE.

Level of evidence III.

Keywords Cryoablation · Paravertebral block · Nuss · Pectus excavatum · Opioid · Length of stay

Introduction

Pectus excavatum is the most common congenital chest wall anomaly in children [1–3]. Minimally invasive pectus excavatum repair (MIRPE) using the Nuss procedure is the procedure of choice for symptomatic patients [4, 5]. Intercostal nerve cryoablation (INC) has been a paradigm shift for pain control in patients undergoing MIRPE. Incorporation of INC has resulted in decreased postoperative pain, postoperative opioid requirement, and length of stay compared to previously reported methods such as opioid patient-controlled analgesia, thoracic epidural, or paravertebral blocks (PVB) [6–8]. The incorporation of INC into clinical practice has been reported as monotherapy or combined with adjuncts such as single-shot PVB, intercostal nerve blocks

This study was presented at the Minnesota Surgical Society, Fall Conference 2023. There are no other disclosures.

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(INB), and incisional local anesthetic infiltration [7]. Postoperatively, patients pain control regimens extend beyond INC, most commonly accompanied by a multimodal pain management approach consisting of both opioid and non-opioid analgesics like acetaminophen, ibuprofen, and gabapentin [6].

Postoperative pain scores, opioid requirement, and length of stay are common outcomes reported in the literature on MIRPE with INC [6–11]; however, the impact of INC on the number of individual medications children require following MIRPE is unknown. In the setting of multimodal pain control along with management of symptoms like nausea and constipation, patients may require a significant number of individual medications during their treatment course. A greater number of medications per day has been associated with increased morbidity and length of stay in adults undergoing surgical procedures [12]; however, this has not been well studied in children. The purpose of this study is to report the overall medication requirement following MIRPE with INC and INB (INC-INB) in addition to opioid requirement and length of stay. Patients receiving INC-INB will be compared to a historical cohort of patients who underwent placement of paravertebral catheters (PVB). We hypothesize that compared to PVB, INC-INB would be associated with decreased overall postoperative medication requirement in addition to decreased opioid requirement and length of stay.

Methods

Cohort

After approval by our Institutional Review Board, a retrospective cohort study, identifying patients ≤ 20 years of age who underwent MIRPE using the Nuss procedure was conducted. Those who underwent MIRPE with INC-INB were compared to a group of historical controls who underwent MIRPE with continuous bilateral percutaneous PVB infusions from January 2018 to March 2023. Patients who underwent revision following primary pectus repair were excluded. The procedures were performed by the same group of surgeons throughout the study period, with no change in institutional clinical management guidelines.

Data collection

Data collected included demographics, Haller index, operative details (analgesic technique, bar fixation technique, number of bars, and duration of procedure), and clinical outcomes including length of stay (LOS), opioid use, benzodiazepine use, and number of unique medications.

Opioid quantity was calculated as Oral Morphine Equivalent (OME) in milligrams per kilogram body weight (OME mg/kg) [13]. Unique medications included all in-hospital medications except for intravenous (IV) fluids and preexisting outpatient prescriptions.

Analysis and outcomes

Statistical analyses were performed using IBM SPSS statistical software version 28.0 (IBM Corp., Armonk, NY) and R Studio (R Core Team 2024 Vienna, Austria). The primary outcome was the number of unique postoperative medications administered, and secondary outcomes included postoperative opioid requirement and length of stay. Continuous and categorical variables were compared using Kruskal-Wallis and Chi-square tests, respectively. A p -value < 0.05 was considered statistically significant. Data is presented as mean, median, and frequency with percentages.

Surgical technique

All patients underwent induction of general anesthesia and double-lumen endotracheal tube placement. Patients in the historical cohort then underwent placement of bilateral percutaneous PVB under ultrasound guidance at thoracic levels T2-T3, T3-T4, or T4-T5 for continuous infusion of 1% lidocaine. Unilateral thoracoscopy was then performed for bar placement. Patients in the INC-INB cohort underwent bilateral sequential thoracoscopy with cryoablation of intercostal nerves 3–7 to -60 degrees Celsius for 120 s each using the cryosphere probe. Each cryoablation site was then anesthetized with local anesthetic at weight-based dosing per surgeon choice. Bar placement and securement were then performed from the side per the surgeon's discretion.

Postoperative management

Throughout the study period, scheduled acetaminophen and ketorolac followed by ibuprofen were used postoperatively. Adjunctive opioids, benzodiazepines, gabapentin, laxatives, and anti-emetic medications were available to patients and used as needed. Gabapentin was not preemptively used. In patients in the historical PVB cohort, adjunctive pain medications were started at the discretion of the Pediatric Pain service in collaboration with the Pediatric Surgery service per our institutional practice management agreement. Pain management of patients who underwent INC-INB was performed by the Pediatric Surgery service. Ambulation was strongly encouraged as early as postoperative day zero. Patients in the historical cohort were discharged with PVB in place which were removed at home. Prescriptions were provided routinely for acetaminophen and ibuprofen with

other medications prescribed only if patients required them during their postoperative stay.

Results

Cohort characteristics

Of 164 patients who underwent MIRPE, 74 (45%) had PVB placed and 90 (55%) underwent INC-INB. The mean $\pm$ sd age was 14.8 ± 1.8 years and 126 (76.8%) patients were male. The average Haller index was 4.2 ± 1.1 and two bars were required in 7% ($n=11$) of patients. Of the 90 patients who underwent INC-INB, 46 (51%) received 0.25% bupivacaine with epinephrine, 25 (28%) received liposomal bupivacaine, 15 (17%) received 0.5% bupivacaine with epinephrine, and 4 (4%) received a combination of 0.5% bupivacaine and liposomal bupivacaine. There were no differences in demographics, Haller index, or use of a second bar between PVB and INC-INB cohorts (all $p > .05$; Table 1).

Postoperative medication requirement

Patients who underwent INC-INB received fewer median (IQR) total postoperative medications compared to those who received PVB [6 (4, 7) vs. 12 (11, 13), $p < .001$], (Fig. 1). While 93% ($n=69$) of patients with PVB received

opioids postoperatively in the hospital, only 43% ($n=39$) of INC-INB patients received opioids, with lower median (IQR) consumption [1.6 (0.7, 3.0) vs. 0 (0, 0.2) OME/kg, $p < .001$] (Fig. 2). Homegoing opioid prescriptions were needed in fewer patients who underwent INC-INB than PVB (27% vs. 76%, $p < .001$), resulting in less amount of opioid being prescribed for outpatient use in the INC-INB cohort [1.1 (0.8, 1.8) vs. 3.9 (2.6, 5.6) OME/kg, $p < .001$] (Fig. 2). Among those who received postoperative opioid prescriptions, the need for refill was similar between the two groups (29% vs. 16%, $p = .18$). No patients who underwent INC-INB received gabapentin as an inpatient, while 48 (64.9%) patients with PVB received gabapentin ($p < .001$). Outpatient gabapentin within one year was prescribed to 4 (4%) patients who underwent INC-INB compared to 19 (26%) patients with PVB ($p < .001$).

Surgical outcomes

While median (IQR) operative time was greater when INC-INB was used compared to PVB [110 (102, 122) vs. 69 (60, 82) minutes, $p < .001$], total time in the operating room was similar [185 (169, 199) vs. 193.5 (174, 210) minutes, $p = .10$]. Median LOS was shorter in INC-INB patients compared to PVB [30 (28, 34) vs. 59 (55, 77) hours, $p < .001$]. Regarding complications, one patient required placement of a chest tube and there were two surgical site infections. Other postoperative complications were minor, and overall complications were rare and not different between cohorts [3 (3%) vs. 5 (7%), $p = .31$].

Discussion

The minimally invasive Nuss procedure is the standard of care for surgical correction of pectus excavatum in children. Postoperative pain was historically managed primarily with opioids, evolving over time to include multimodal pain management strategies that emphasize non-opioid medications. The addition of regional adjuncts such as epidural catheters, intercostal nerve blocks, and paravertebral blocks has further improved the management of postoperative pain. Since INC was first reported for pain control in the Nuss procedure, there has been widespread adoption in the pediatric MIRPE population. Here we report that, in addition to decreased opioids, INC with INB is associated with a 50% reduction in the number of medications children are exposed to postoperatively when compared to children who received PVB.

Between medications to control pain and manage other symptoms that can occur postoperatively, children undergoing pectus excavatum repair may be exposed to a large

Table 1 Baseline cohort characteristics

	Overall <i>n</i> = 164	Intercostal cryoablation <i>n</i> = 90 (55%)	Paraverte- bral block <i>n</i> = 74 (45%)	<i>p</i> -value
Gender				0.67 ¹
Male	38 (23%)	68 (76%)	58 (78%)	
Female	126 (77%)	22 (24%)	16 (22%)	
Age	15 (14, 16)	15 (14, 16)	15 (13.3, 16)	0.51 ²
Haller Index ^a	3.8 (3.6, 4.3)	3.8 (3.6, 4.3)	3.8 (3.6, 4.4)	0.46 ²
Presence of second bar	11 (6.7%)	5 (5.6%)	6 (8.1%)	0.52 ¹
Total time in operating room (min)	189 (171, 202)	185 (169.3, 199)	194 (174, 209.6)	0.096 ²
Total operating time (min)	95 (69, 112)	110 (102, 122)	69 (60, 81.5)	<0.001 ²
Length of Stay (hours)	51 (30, 58)	30 (28, 34)	59 (55, 77)	<0.001 ²
Discharge POD One	72 (44%)	71 (79%)	1 (1%)	<0.001 ¹

^a Based on inspiratory chest radiograph

^{b1} Chi Square Test; ² Kruskal Wallis Test

^c Abbreviations: POD – Post operative day

^d Reported as median (Interquartile Range) or frequency (percent)

Fig. 1 Total number of individual postoperative medications administered for patients receiving paravertebral blocks (PVB) versus intercostal cryoablation with intercostal nerve block (INC-INB)

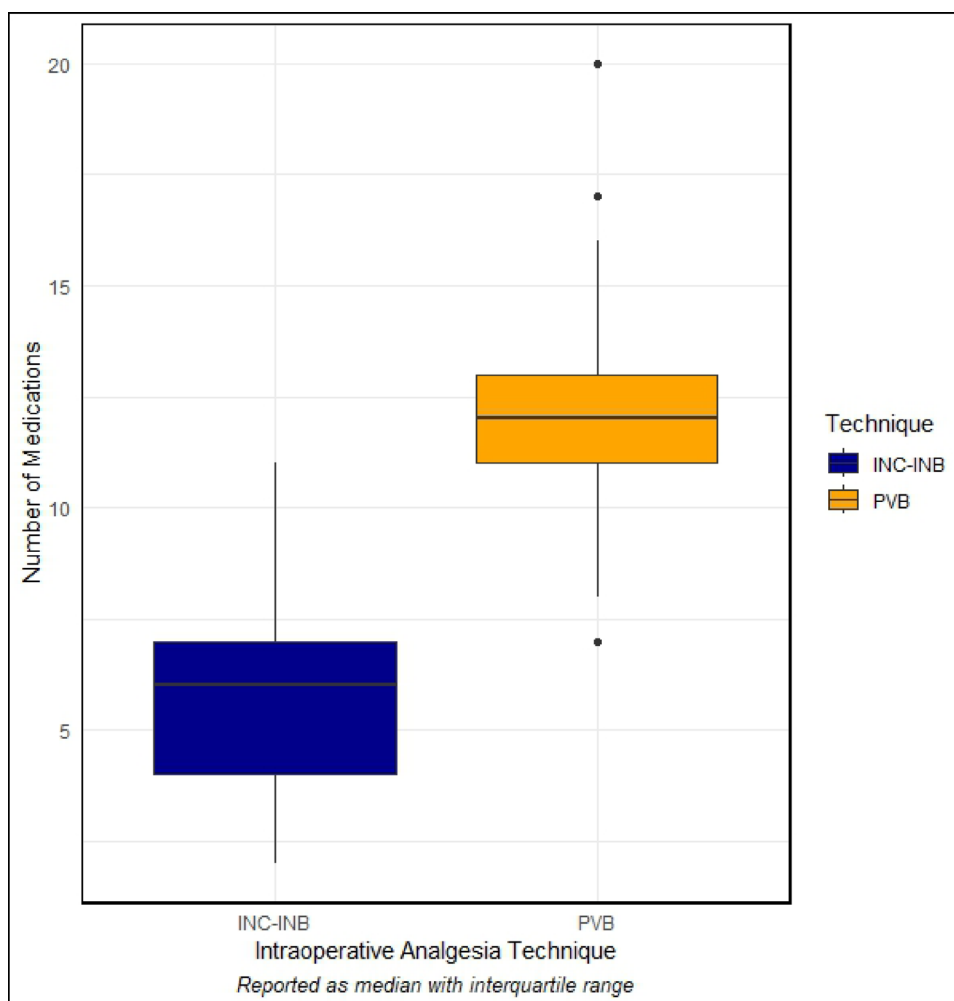
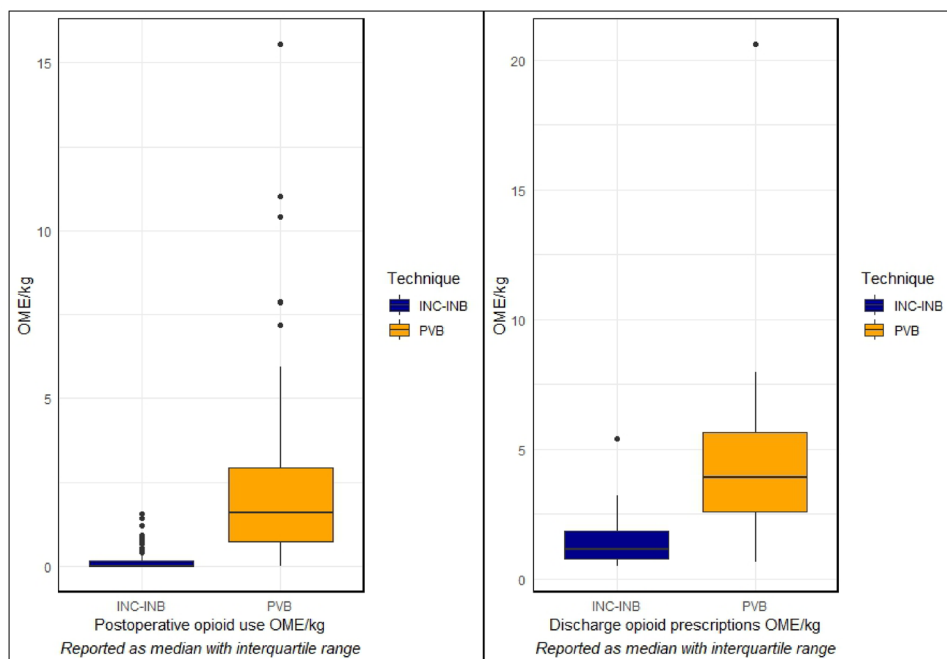


Fig. 2 Comparison of opioid requirement postoperatively and at discharge (in OME/kg) based on intraoperative analgesic method of paravertebral blocks (PVB) or intercostal cryoablation with intercostal nerve block (INC-INB)



number of pharmacologic agents. Opioid-related side effects, such as nausea, vomiting, constipation, and urinary retention, often require management with the addition of other medications [10]. Epidural and paravertebral analgesia can be helpful in reducing systemic opioid requirement; however, are associated with postoperative nausea and are limited in their duration of analgesia [14]. By eliminating the need for these adjuncts and minimizing opioid need, INC, directly and indirectly, reduces the need for medications. This likely contributes to the shortened length of stay associated with INC, as the number of unique medications one is exposed to has been associated with length of stay in adult surgical cohorts [12]. Minimizing polypharmacy while maintaining excellent, opioid-sparing pain control and short length of stay should be a goal for the postoperative management of children who undergo MIRPE.

Gabapentin is a non-opioid medication that is reported as part of multimodal pain management strategies for MIRPE, including as an adjunct to INB and INC [11, 18]. Initially used as an anticonvulsant and antidote for chronic, neuropathic pain, its use for acute neuropathic pain has steadily increased. In the case of MIRPE, it is used to alleviate intercostal neuropathy related to the procedure itself. While individual studies demonstrate the efficacy of perioperative gabapentin in improving postoperative pain in adults, these benefits do not hold in meta-analyses [15]. Additionally, gabapentin resulted in a greater risk of dizziness and visual disturbance without a reduced risk of developing chronic pain postoperatively [19]. A recent review of 11 randomized controlled trials and four retrospective studies designed to specifically assess perioperative gabapentin in children concluded there is insufficient data to support routine use [16]. In this study, the opioid requirement was minimal, and the length of stay was approximately one day with rare gabapentin use, supporting the use of INC-based pain management strategies without gabapentin. Further studies are needed to specifically evaluate the role of gabapentin for patients undergoing MIRPE that include longer-term outcomes such as duration of gabapentin use and chronic pain.

This study reports INC accompanied by INB, which is one of many adjuncts to INC reported. Cryoablation results in axonal degeneration which can take up to 24 h to reach maximum efficacy, so in theory, INB may improve pain control during this initial time period [20]. INB is an appealing adjunct as it can be performed by the surgeon expeditiously with the same exposure needed for INC as opposed to INC with adjunctive paravertebral analgesia which has also been reported [10, 11, 18]. Same-day discharge has also been reported by Rettig et al. following MIRPE with both INC and INB [10]. INB alone has been described and was associated with decreased opioid requirements compared to a patient-controlled analgesia-based approach; however, the

median length of stay was 3 days which is greater than what is reported for contemporary INC-based approaches [17]. INB does carry a small risk of bleeding from adjacent intercostal vessels and adds time, albeit small, to the operation [21]. A comparison of INC alone to INC with INB is needed to confirm the utility of this adjunct.

This study was a single-institutional retrospective study using historical controls for comparison aimed at specifically evaluating postoperative polypharmacy between a cryoablation-based and non-cryoablation analgesic approach and, therefore, has limitations. Use of a historical comparison cohort is inherently limited due to changes, often intangible, that occur over time to improve care. PVB anchored our previous analgesic approach so it is not known if findings would be similar if the comparison group had other approaches such as thoracic epidural. Since patients with PVB were followed by the Pediatric Pain Service and those with INC were not, it is possible that this management difference confounded outcomes. It is also possible that patients received care elsewhere postoperatively and we could have missed complications or opioid prescriptions, though one-month follow-up occurred in all patients. Lastly, long-term outcomes were not assessed prospectively.

In conclusion, INC with INB is associated with decreased polypharmacy following MIRPE in children, in addition to decreased opioid requirement and shorter length of stay. The number of unique medication exposures should be considered when evaluating postoperative pediatric surgical care in the modern era. The utility of adjuncts to INC such as INB and gabapentin requires further, dedicated study in this population.

Author contributions CRediT Authorship Statement: M.T. P: Data curation, Formal Analysis, Writing – review and editing, Visualization, Critical review, prepared Figs. 1 and 2 and S. R. R: Conceptualization, Methodology, Data curation, Writing – original draft ; (A) G: Formal Analysis, Data curation, Critical review; D. D. P Jr.: Conceptualization, Methodology, Data curation, Critical review; D. (B) K: Conceptualization, Methodology, Critical review; S. F. P: Administration, Conceptualization, Methodology, Critical review.

Funding The authors did not receive support from any organization for the submitted work.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

Ethical approval This retrospective chart review study was conducted in accordance with the ethical standards of the institutional and national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. The Human In-

vestigation Committee (IRB) of Mayo Clinic approved this study.

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